

- Teacher's Class

1. normal
 2. infected
 3. supernatant
 4. stimulated
 5. unstimulated

using super()
super() is used to call the Parent class constructor or method.

```
class Person:
    def __init__(self, name):
        self.name = name
```

```
class Student(Person):
    def __init__(self, name, role):
        super().__init__(name)
        self.role = role
```

```
s = Student("Krish", 101)
print("Name:", s.name)
print("Role:", s.role)
```

```
# Name: Krish
# Role: 101
```

Ques why do we use super?

- Ans
- ① Reusing parent code.
 - ② Avoiding rewriting the same code.
 - ③ Proper initialization of inherited attribute.

Exercise: Mini library management system.

- Person
- Student
- Teacher
- Different borrowing limit.



```
class child1(Parent):  
    pass
```

```
class child2(Parent):  
    pass
```



Method Overriding
When child class changes the Parent class
Method

```
class Animal:
```

```
    def sound(self):  
        print("Animal make sound")
```

```
class Dog:
```

```
    def sound(self):  
        print("dog barks")
```

```
d=Dog()
```

```
d.sound()
```

dog barks.

Parent class → Base class

Child class → derived class

class child(father, mother):

Pass.

↓

Pass in python is a null statement (python does not allow empty box so we use Pass)
we can use inside → function, classes, loop, statements

C = child()

C.speak()

C.speak2()

Driving
Cooking

(3) Multilevel Inheritance (Grandparent - parent - child)

class Grandparent:

def house(self):

print("have a house")

class Parent(Grandparent):

~~print("have a car")~~

def car(self):

print("have a car")

class child(Parent):

def bike(self):

print("have a bike")

(4) Hierarchical Inheritance (one parent multiple child)

class Parent:

def property(self):

print("parent property")

Inheritance is to

It is an OOPS concept where one class can get the properties and method of another class (parent).

- It helps in:
- ① Code reusability
 - ② Reducing duplication
 - ③ Creating a logical relationship between classes.

There are 4 types of inheritance.

① Single inheritance (One child inherits from 1 parent)

class Animal:

```
def eat(self):  
    print("Eating")
```

class Dog (Animal):

```
def bark(self):  
    print("barking")
```

② Multiple inheritance. (One child from multiple parent) ^{inherit}

class Father:

```
def skill1(self):  
    print("driving")
```

class Mother:

```
def skill2(self):  
    print("cooking")
```